Who Pays and Where?

Local Housing Market Adjustments to Immigration in Spain

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Outline

Research Question

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Does immigration at the municipal level in Spain raise housing prices ("who pays?")? Where do these effects concentrate across the urban hierarchy, and how do they evolve over time ("where?")? Which local adjustment channels — housing supply, native mobility, local economic activity — absorb or amplify the demand shock?

Introduction and Motivation

National Context I: Housing Supply & Prices

Figure: New construction permits

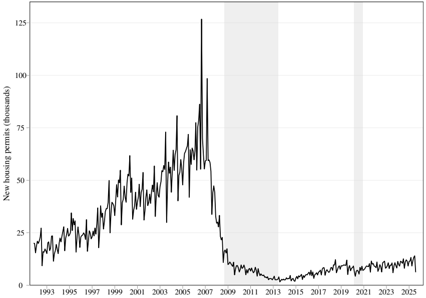
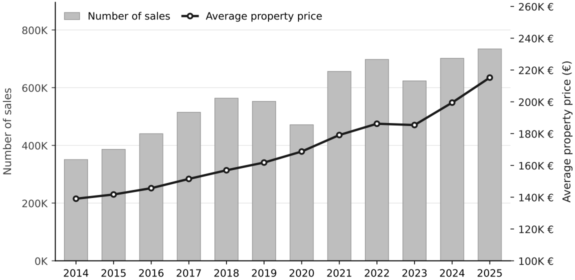


Figure: Transaction Prices and Volume, Spain



- Supply (building permits) remained **far below pre-crisis levels** throughout the period.
- Average transaction prices **increased** $\sim 54\%$ throughout the analysis period.
- In 2025, the volume of sales **more than doubled** the 2014 levels.

National Context II: Immigration Dynamics

Figure: Foreign born share in the main EU receiver countries

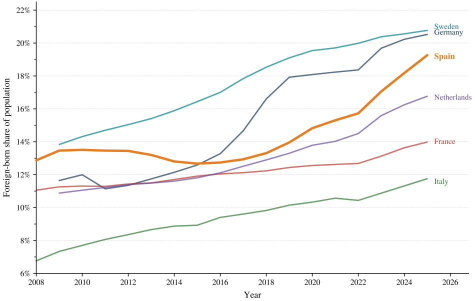
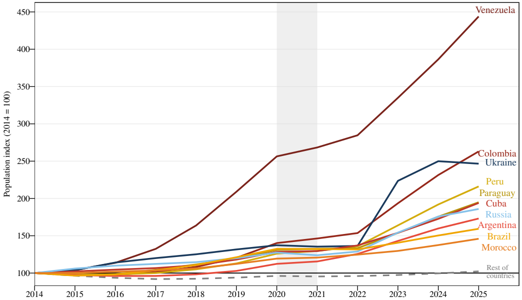


Figure: Inflows by main origins, Spain



- Foreign-born share rose from $\sim$12.8% in 2014 to $\sim$19.3% in 2025.
- Non-EU inflows dominated, led by **Venezuela, Colombia, Peru, and Paraguay, alongside Morocco**.
- Roughly $\sim$80% are settled in the biggest $\sim$ 10% municipalities.

Municipal Context: Immigration & Price Changes, 2014–2025

Figure: Immigrant growth, 2014-2025

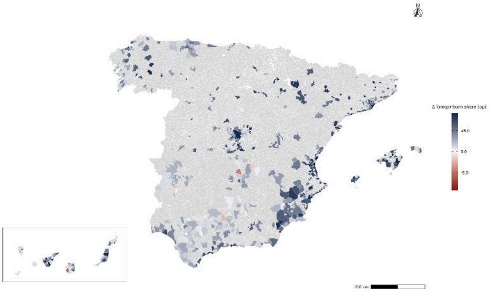
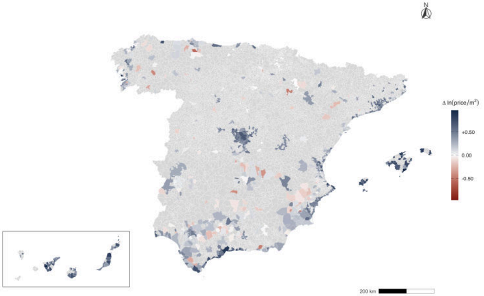


Figure: Housing Price Growth, 2014–2025



- Both immigration and price increases are **spatially concentrated** in coastal and metropolitan areas.

Hypotheses

H1 — Short-run price pressure

Immigration exerts upward pressure on housing prices in the short run when local adjustment channels are insufficient to absorb the demand shock.

H2 — Accumulation over time

Under persistent supply constraints and limited spatial mobility, the price effect of immigration accumulates: long-run effects exceed short-run estimates.

H3 — Heterogeneity across the urban hierarchy

The price effect varies systematically with the degree of urbanization: local supply capacity and labor market conditions determine whether demand pressure dissipates or accumulates in the long run.

Summary of Results

Main Findings

1. **Large, causal price effects**: a 1pp increase in the net immigration rate raises local housing prices by $\sim$0.73% in the short run, accumulating to $\sim$2.25% in the long run (2014–2025).
2. **No aggregate native response**: natives neither systematically leave nor concentrate in high-immigration municipalities on average.
3. **Inelastic supply**: no significant response in construction, urban land, or residential use over a decade.
4. **Urbanization matters**: within non-urban municipalities the long-run effect reaches $\sim$4pp, while the within-urban long-run effect is no longer statistically significant.
5. **Immigration dynamises non-urban economies**: firm creation and income gains induce native co-location *within non-urban municipalities*, compounding demand pressure in the long run.

Data

Data: Variables, Sample & Coverage

Sample: **699 municipalities** >10k inhabitants[1], 2014–2025.

- **Outcomes:** $\Delta \log(\text{price})$, $\Delta \log(\text{price}/m^2)$ (Portal Estadístico del Notariado).
  - *Robustness:* $\Delta \log(\text{appraisal value})$ and rent price index yield similar positive and significant results.
- **Treatment:** Δ Immigrant share (Padrón Municipal 2014–2022 & Censo Anual de Población 2023–2025).
- **Instruments:** Bartik shift-share; spatial gravity shifter.
- **Controls:** *Demand-side*: $\Delta \log$ affiliated workers (TGSS), Bartik employment demand index (2011 Census sectoral shares $\times$ national sectoral employment growth). *Supply-side*: $\Delta \log$ construction firms (DIRCE), $\Delta \log$ residential land area, $\Delta \log$ total urban area, $\Delta \log$ hospitality/tourist-use share (all Catastro).

▸ Sample coverage

[1] Excl. País Vasco & Navarra due to data limitations.

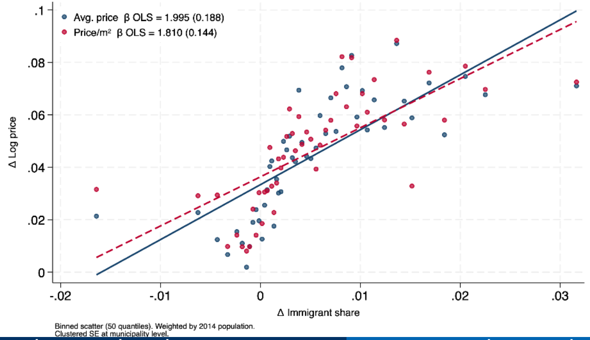
Estimation Strategy

Naïve Specification

Baseline OLS

$$\Delta \log P_{c,t} = \alpha + \beta \frac{\Delta \mathrm{M}_{c,t}}{Pop_{c,t-1}} + \varepsilon_{c,t}$$

Figure: Immigration and Housing Prices: Raw Correlation



Problem

OLS estimates are subject to three sources of **bias**: **omitted local shocks**, **endogenous location choice** of immigrants, and **spatial dependence** across municipalities.

Problem I: Supply & Socioeconomic Confounders

- **Supply shocks**: local construction booms can simultaneously attract workers and reduce prices, therefore causing omitted variable bias.
- **Labour market**: local employment growth attracts both immigrants and natives, inflating prices independently.
- **Amenity trends**: municipalities with improving amenities attract residents of all origins.

Solution

Two-way fixed effects (year + province), province linear trends, demand-side controls ($\Delta \log$ affiliated workers (TGSS), Bartik employment demand index), and supply-side controls ($\Delta \log$ construction firms, residential land area, total urban area, hospitality share):

$$B_{c,t} = \sum_s \lambda_{c,s,2011} \cdot \Delta \log E_{s,t}$$

where $\lambda_{c,s,2011}$ is municipality c's employment share in sector s in 2011 (Spanish Population Census), and $\Delta \log E_{s,t}$ is the national employment growth in sector s.

Problem II: Endogenous Location Choice — Settlement Shares

$$\theta^{(\tau)}_{c,2005} = \frac{M^{\tau}_{c,2005}}{\sum_c M^{\tau}_{c,2005}}$$

- For each country of origin τ and municipality c: what fraction of origin τ's total stock in Spain lived in municipality c in base year 2005?
- Predetermined w.r.t. the estimation period (2014–2025) — reflects historical networks, not current housing conditions.
- Source: *Padrón Municipal*, INE. 28 origin countries tracked.

List of countries

Problem II: Endogenous Location Choice — Shift-Share Instrument

$$\Delta\widehat{M}_{c,t} = \sum_{\tau} \theta^{(\tau)}_{c,2005} \cdot \Delta M^{\tau}_{t}$$

- **Predictor**: allocates national inflows ΔM^{τ}_{t} (by origin τ) to municipality c using predetermined 2005 settlement shares.
- **Predicted population**: $\widehat{Pop}_{c,t-1} = \widehat{M}_{c,t-1} + \widehat{N}_{c,t-1}$, where $\widehat{N}_{c,t} = \frac{N_{c,2005}}{\sum_c N_{c,2005}} \cdot \Delta N_t$ is the analogous predicted native population — avoiding mechanical correlation between the denominator and contemporaneous local conditions.
- **First stage**: $\frac{\Delta M_{c,t}}{Pop_{c,t-1}} = \alpha + \pi \frac{\Delta\widehat{M}_{c,t}}{\widehat{Pop}_{c,t-1}} + \phi \Delta X^{D}_{c,t} + \Phi \Delta X^{H}_{c,t} + \lambda_t + \lambda_r + (\lambda_t \cdot \lambda_r) + \varepsilon_{c,t}$
- **Exclusion**: $\Delta\widehat{M}_{c,t}$ is orthogonal to local housing demand shocks.
- **Follows**: Card (2001), Goldsmith-Pinkham et al. (2020), Borusyak et al. (2022).

▸ Gravity Push

Instrument Validity I: First Stage & Relevance

Dep. Variable: $\Delta\text{Immigration}_{c,t}$	(1)	(2)	(3)	(4)
Panel A: Short-Run (Annual Differences)				
$\widehat{\Delta\text{Immigration}}_{c,t}$	0.603***	0.604***	0.599***	0.752***
	(0.057)	(0.058)	(0.057)	(0.052)
KP F-stat	111.9	109.6	109.0	**208.6**
R-Squared	0.584	0.585	0.587	0.671
Clusters	699	699	699	699
Observations	7,689	7,689	7,689	7,689
Panel B: Long-Run (Long Difference 2014–2025)				
$\widehat{\Delta\text{Immigration}}_{c,t}^{14-25}$	0.505***	0.499***	0.470***	0.605***
	(0.142)	(0.145)	(0.145)	(0.069)
KP F-stat	47.5	42.9	39.6	**77.7**
R-Squared	0.337	0.349	0.377	0.623
Clusters	46	46	46	46
Observations	699	699	699	699
Year FE	Yes	Yes	Yes	Yes
Housing Supply Controls	No	Yes	Yes	Yes
Demand Controls	No	No	Yes	Yes
Province FE	No	No	No	Yes
Province $\times$ Trend	No	No	No	Yes

All specifications weighted by 2014 municipal population. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Problem III: Spatial Spillovers — Exogenous Spatial Pull

$$\mathrm{Pull}_{c,t}^{\mathrm{exog}} = \frac{\displaystyle\sum_{j \neq c} \frac{A_j}{d_{cj}^2} \Delta \widehat{M}_{j,t}}{\displaystyle\sum_{j \neq c} \frac{A_j}{d_{cj}^2}}$$

where $\Delta\widehat{M}_{j,t}$ is the shift-share-predicted immigrant flow in neighbor j, d_{cj} the geodesic distance to c, and A_j the area of j

- **Spatial effect**: immigration in nearby municipalities affects local housing demand — ignoring this induces spatial autocorrelation in residuals.
- **Construction**: inverse-square distance kernel weights nearby municipalities more; the area term corrects for larger municipalities mechanically attracting more immigrants. The denominator normalises $\mathrm{Pull}_{c,t}^{\mathrm{exog}}$ as a weighted *average*, comparable across municipalities regardless of geographic isolation.
- **Role**: enters specification as an additional control, purging omitted spatial demand pressure so that β captures the causal effect of the *own* immigration shock, holding the neighbours' pressure constant.

Full Specification

Preferred 2SLS Specification

$$\Delta \log P_{c,t} = \alpha + \beta \frac{\Delta \mathrm{M}_{c,t}}{Pop_{c,t-1}} + \phi \Delta X^{D}_{c,t} + \Phi \Delta X^{H}_{c,t} + \lambda_t + \lambda_r + (\lambda_t \cdot \lambda_r) + \varepsilon_{c,t}$$

$\frac{\Delta M_{c,t}}{Pop_{c,t-1}}$: net immigration rate, instrumented with $\frac{\Delta \widehat{M}_{c,t}}{\widehat{Pop}_{c,t-1}}$.

$\mathrm{Pull}^{\mathrm{exog}}_{c,t}$ enters as an additional control in the preferred specification.

Identification

β captures the causal semi-elasticity of housing prices with respect to a one-percentage-point increase in the net immigration rate, conditional on local demand and supply conditions, spatial spillovers, and time-invariant provincial trends.

Results

Immigration and Housing Prices: Main Results

Alternative outcomes

Dep. Variable: Δ log Avg. Transaction Price	(1)	(2)	(3)	(4)
Panel A: OLS				
ΔImm.	1.159***	1.129***	1.057***	0.522***
	(0.217)	(0.216)	(0.206)	(0.137)
Panel B: 2SLS				
ΔImm.	2.169***	2.142***	2.061***	0.840***
	(0.438)	(0.423)	(0.407)	(0.259)
KP F-stat	111.9	109.6	109.0	208.6
Panel C: 2SLS + Exogenous Spatial Pull				
ΔImm.	2.144***	2.116***	2.041***	0.727***
	(0.426)	(0.413)	(0.400)	(0.265)
Δ Spatial Pull	-0.060***	-0.062***	-0.054***	-0.067***
	(0.023)	(0.021)	(0.020)	(0.016)
KP F-stat	116.2	114.1	112.7	212.8
Year FE	Yes	Yes	Yes	Yes
Supply Controls	No	Yes	Yes	Yes
Demand Controls	No	No	Yes	Yes
Province FE & Trend	No	No	No	Yes
N	7,689	7,689	7,689	7,689

Weighted by 2014 population. Standard errors clustered at the municipal level in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Dynamic Results: Long Differences

Long differences: we stack differences over $h \in \{2,3,4\}$ years and a single cross-section spanning 2014–2025. Longer differences better capture the **full local market adjustment** to immigration shocks.

2SLS	$h=2$ (1)	$h=3$ (2)	$h=4$ (3)	2014–2025 (4)
Panel A: Baseline				
Δ^hImm.	1.335***	1.852***	2.185***	2.444***
	(0.277)	(0.331)	(0.414)	(0.796)
Panel B: + Exogenous Spatial Pull (preferred)				
Δ^hImm.	1.242***	1.799***	2.137***	2.246***
	(0.276)	(0.328)	(0.409)	(0.790)
Δ^h Spatial Pull	-0.051***	-0.029**	-0.024	-0.066***
	(0.014)	(0.014)	(0.015)	(0.021)
Year FE & Controls	Yes	Yes	Yes	No
Province FE	Yes	Yes	Yes	Yes
Province $\times$ Trend	Yes	Yes	Yes	No
KP F-stat	204.5	204.6	181.0	75.9
N	6,990	6,291	5,592	699

Standard errors clustered at the municipal level in columns (1)–(3) and at the province level in column (4). Weighted by 2014 population. $^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$.

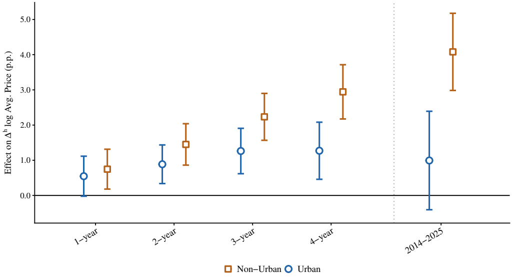
Adjustment Channels: Native Mobility, Supply & Local Economy

Dep. variable (Δ^h)	*1 year* (1)	$h = 2$ (2)	$h = 3$ (3)	$h = 4$ (4)	*2014–2025* (5)
Panel A: Demographic response					
Native population	0.036	0.030	0.027	0.012	−0.020
	(0.063)	**(0.070)**	**(0.084)**	**(0.109)**	**(0.140)**
Panel B: Supply-side response					
Log urban area	−0.215	−0.288	−0.233	−0.321	0.471
	(0.189)	**(0.190)**	**(0.210)**	**(0.274)**	**(0.614)**
Residential land share	0.020	0.027	0.037	0.045	0.045
	(0.023)	**(0.025)**	**(0.030)**	**(0.037)**	**(0.068)**
Log construction firms	−0.301	−0.076	−0.337	−0.415	−0.436
	(0.385)	**(0.354)**	**(0.417)**	**(0.470)**	**(1.198)**
Hospitality land use share	0.319	0.651	0.860	0.662	1.63
	(0.976)	**(0.907)**	**(0.935)**	**(1.077)**	**(2.389)**
Panel C: Local economic activity					
Log affiliated workers	0.048	0.108	0.171	0.205	0.579
	(0.180)	**(0.180)**	**(0.195)**	**(0.236)**	**(0.428)**
Log total firms	0.399***	0.538***	0.688***	0.884***	1.212***
	(0.111)	**(0.098)**	**(0.105)**	**(0.122)**	**(0.149)**
Log service firms	0.081	0.166	0.271*	0.399**	0.351
	(0.149)	**(0.137)**	**(0.144)**	**(0.163)**	**(0.289)**
Log median income	0.469***	1.378***	—	—	1.473***
	(0.128)	**(0.025)**			**(0.433)**

2SLS, full control set, weighted by 2014 population. Clustered SE at municipal level in (1)–(4), province level in (5). Income long diff. covers 2020–2023. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Heterogeneity: Urban Degree

Figure: Dynamic Causal Effects of Immigration on Housing Prices by Degree of Urbanization



- **Within-group identification**: estimates compare high- vs. low-immigration municipalities *within* each urban category.
- The total long-run non-urban effect reaches $\sim$**4.08pp**; the within-urban long-run effect is **no longer significant**.

▸ Urban hierarchy (SR) ▸ Full table

Mechanisms I: Native Mobility by Urban Degree

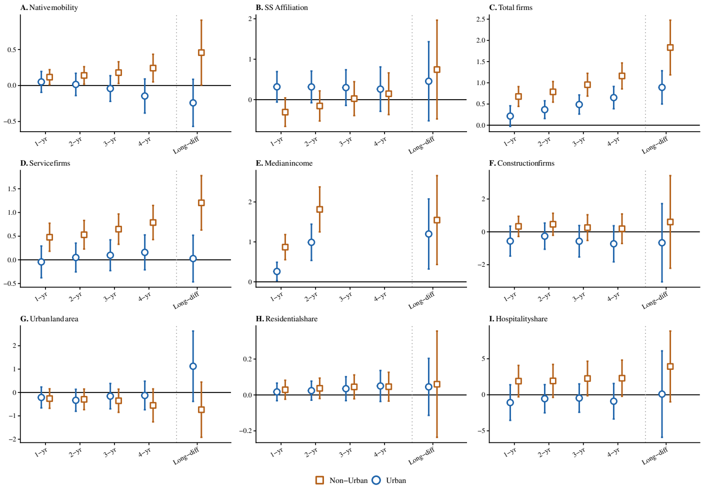
Δ^h Native Mobility	$h = 1$ (1)	$h = 2$ (2)	$h = 3$ (3)	$h = 4$ (4)	*2014–2025* (5)
Urban	0.050	0.015	−0.042	−0.146	−0.241
	(0.074)	(0.079)	(0.091)	(0.121)	(0.167)
Non-Urban	0.066	0.125*	0.221***	0.388***	0.699**
	(0.062)	(0.066)	(0.074)	(0.094)	(0.296)
Controls, FE, Pull	Yes	Yes	Yes	Yes	Yes
N	7,689	6,990	6,291	5,592	699

- Within non-urban municipalities, immigration shocks attract native in-migration, with the effect **emerging gradually** from $h = 2$ and strengthening over time.
- Within urban municipalities, the response is never significant.

▸ Urban hierarchy (SR)

Mechanisms II: Local Adjustment by Urban Degree

Figure: The Effect of Immigration on Other Outcomes by Degree of Urbanization



Robustness

Main results are robust to:

1. **Alternative outcomes**: price/m^2 and appraisal values converge to $\approx 2.0\%$ in the long run; rent index responds positively but attenuated (0.14% short run, 0.41% long run), consistent with staggered tenancy adjustment. Detailed results
2. **Gravity IV**: national shifts predicted from origin-country push factors (lagged GDP, emigration to other destinations) recover 0.70% in the short run and 2.94% in the long run. Detailed results
3. **CRE (Mundlak) correction**: estimates for $h \in \{2, 3, 4\}$ in line with the baseline; Mundlak terms jointly significant, confirming the correction is warranted. Detailed results
4. **Excluding the largest municipalities** (top 5/10/20 by 2014 population): 0.69–0.85% short run, 2.44–2.77% long run. Detailed results
5. **Unweighted regressions**: 0.89% short run, 3.35% long run.
6. **Fixed 2014 denominator** (purging endogenous population changes): 0.72% short run, 2.25% long run.

Discussion

Discussion

Labour & Agglomeration

- Immigration augments **labour supply and wages**.
- **Housing costs absorb** much of the wage premium for incumbent residents.
- Net welfare effect depends on **ownership status**.

Spatial Sorting

- **No aggregate native displacement**; within non-urban municipalities, natives **co-locate** with immigrants.
- Sorting is driven by **economic activation**.
- Pattern is *consistent with* a spatial cascade toward smaller markets, though cross-tier reallocation is not directly identified.

Policy Implications

- Supply inelasticity is the binding constraint — **zoning reform** is the key lever.
- Demand pressure is most persistent in **non-urban** markets — supply policy cannot target only large cities.
- Renters and non-owners bear the **distributional cost**; owners benefit via wealth effects.

Conclusion

Conclusion

We find:

1. **Large causal effects** of immigration on housing prices: 0.73% per 1pp in the short run, accumulating to 2.25% in the long run.
2. **Rigid supply**: no significant response in construction, urban land, or residential use over a decade.
3. **No aggregate native displacement**; within non-urban municipalities, immigration attracts native in-migration through local economic gains.
4. **Within-group heterogeneity**: the long-run effect reaches $\sim$*4pp within non-urban municipalities*, while the within-urban effect loses significance.

- **Who pays?** Everyone — through persistently higher prices, and most intensely in non-urban markets where supply cannot respond and native co-location compounds demand.
- **Where?** Price pressure accumulates where local absorption capacity is weakest. Our estimates compare municipalities *within* each urban tier, so they capture how each tier absorbs its own shock rather than reallocation across tiers.

Who Pays and Where?

Local Housing Market Adjustments to Immigration in Spain

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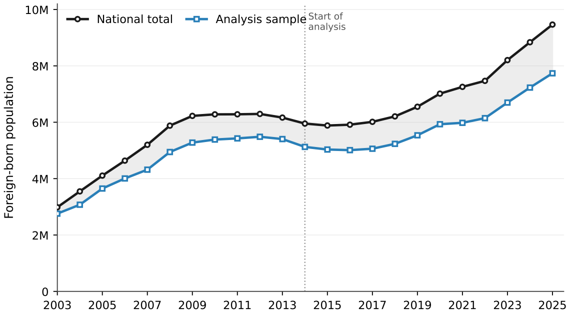
Municipal Immigration: Sample coverage, 2014–2025

Figure: Sample coverage vs. national immigration stock



	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Sample share (%)	86.1	85.5	84.7	84.1	84.3	84.5	84.5	82.4	82.2	81.6	81.7	81.7

Notes: $N = 699$ municipalities. Share is computed as sample population divided by the national total of foreign-born resident population registered in the Padrón Municipal and Censo Anual de Población (INE).

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List of the countries

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Europe		*Africa*		*Asia*	
Germany	Poland	Algeria	Nigeria	China	Pakistan
Bulgaria	Portugal	Morocco	Senegal		
France	Romania				
Italy	Russia				
	Ukraine				
	United Kingdom				
Latin America & Caribbean					
Argentina	Brazil	Colombia	Ecuador	Peru	Uruguay
Bolivia	Chile	Cuba	Paraguay	Dom. Republic	Venezuela

Alternative Housing Outcomes

Robustness

	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	*Long Diff.* (5)
Panel A: Δ^h log *Price per m²* (N = 699 municipalities)					
Δ^hImm.	0.428*	0.770***	1.206***	1.417***	2.020***
	(0.244)	(0.245)	(0.290)	(0.340)	(0.580)
Panel B: Δ^h log *Rent Price Index* (N = 687; long diff. 2014–2023)					
Δ^hImm.	0.140**	0.150**	0.116	0.112	0.409***
	(0.058)	(0.067)	(0.074)	(0.080)	(0.125)
Panel C: Δ^h log *Appraisal Value per m²* (N = 287, >25k inhabitants)					
Δ^hImm.	0.369**	0.723***	0.929***	0.955***	2.034***
	(0.174)	(0.195)	(0.231)	(0.254)	(0.470)

All specifications include the full control set, province FE, province trends (except long diff.) and the exogenous spatial pull; weighted by 2014 population. Clustered SE at municipal level (cols 1–4), province level (col 5). $^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$.

Bartik Instrument — Exogenous National Flows (Gravity Push)

Following Sanchis-Guarner (2023), and addressing concerns regarding Bartik instruments (Borusyak et al., 2022), we predict national shifts from origin-country supply-push fundamentals on a panel of origin countries over 2005–2025:

$$\Delta \widehat{M}_t^{\tau} = \widehat{\beta} \ln \mathrm{GDP}_{\tau,t-1} + \sum_t \widehat{\gamma}_t \cdot \Delta \mathrm{Out}_{\tau} + \widehat{\delta}_{\tau} + \widehat{\gamma}_t$$

- $\ln \mathrm{GDP}_{\tau,t-1}$: lagged origin-country macroeconomic conditions, determining the propensity to emigrate.
- $\Delta \mathrm{Out}_{\tau}$: change in the stock of emigrants from country τ to destinations *other than Spain* (UN DESA, 2024) — a structural supply-push measure, interacted with year FE so its effect varies freely over time.
- $\widehat{\delta}_{\tau}$: origin fixed effects absorb persistent characteristics of each migrant group.
- Fitted values $\Delta \widehat{M}_t^{\tau}$ replace the observed national shifts in the shift-share construction, ensuring the shift component is driven exclusively by origin-country push factors.

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Instrument Validity II: Exogeneity of the Bartik

Back

We construct a forward-looking predictor from 2020–2025 national inflows and regress outcomes measured *before* the shock (2014–2020) on it. If the instrument were correlated with persistent local conditions, we would expect a significant relationship even though future inflows cannot causally affect past outcomes.

	Dependent Variable: Δ Outcomes (2014–2020)						
	$\Delta\log$ Price (1)	Δ Native Pop. (2)	Δ Constr. Firms (3)	Δ Urban Area (4)	Δ Resid. Share (5)	Δ Tourism Share (6)	Δ Affiliates (7)
$\widehat{M}_{c,2020-2025}$	0.771 (0.653)	0.130 (0.103)	0.748 (0.620)	0.747 (0.665)	0.037 (0.040)	0.635 (1.751)	−0.389 (0.274)
Province FE, Controls, Spatial Pull	Yes						
Clusters	46						
Observations	699						

Each column regresses the 2014–2020 change in the outcome on the shift-share instrument constructed from 2020–2025 immigrant flows. All seven coefficients are statistically insignificant: no pre-existing trends in housing prices, demographics, or supply factors. Weighted by 2014 population. SE clustered at the province level. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Gravity IV Robustness: Full Estimates

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Dep. Variable: Δ^h log Avg. Price	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	*Long Diff.* (5)
Δ^hImm.	0.699*	1.196***	1.696***	1.995***	2.938**
	(0.379)	(0.366)	(0.382)	(0.468)	(1.153)
First-Stage Gravity IV	0.644***	0.645***	0.645***	0.628***	0.520***
	(0.054)	(0.054)	(0.050)	(0.051)	(0.101)
KP F-stat	144.1	142.4	165.2	152.3	26.4
Clusters	699	699	699	699	46
N	7,689	6,990	6,291	5,592	699

Δ^hImm. is instrumented with the gravity-based predictor constructed from origin-country supply-push components replacing the observed national shifts in the baseline shift-share instrument. Columns (1)–(4) include province FE and province-specific linear trends; column (5) includes province FE only. All specifications include demand and supply controls, the exogenous spatial pull, and are weighted by 2014 population. SE clustered at municipal level in (1)–(4), province level in (5). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Correlated Random-Effects (CRE) Correction: Full Estimates

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Dep. Variable: Δ^h log Avg. Price	$h = 1$ (1)	$h = 2$ (2)	$h = 3$ (3)	$h = 4$ (4)
Δ^hImm.	0.599	1.250***	2.347***	3.457***
	(0.379)	(0.381)	(0.566)	(0.977)
KP F-stat	109.8	91.2	76.2	56.2
Mundlak joint test (χ^2)	22.29	49.29	79.75	100.33
Clusters	699	699	699	699
N	7,689	6,990	6,291	5,592

Following Mundlak (1978) and Lin & Wooldridge (2019), municipality-level time averages of all time-varying controls (supply, demand, spatial pull, and the immigration rate) are added to each specification; in the linear IV case this correction reproduces the FE-2SLS estimator. The Mundlak joint test rejects the null of zero Mundlak terms across all specifications, confirming the correction is warranted. All specifications include province FE, province-specific linear trends, demand and supply controls, and are weighted by 2014 population. SE clustered at the municipal level. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Additional Robustness: Full Estimates

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Dep. Variable: $\Delta^h \log$ Avg. Price	$h = 1$ (1)	$h = 2$ (2)	$h = 3$ (3)	$h = 4$ (4)	*Long Diff.* (5)
Panel A: Excluding top-5 municipalities by 2014 population					
Δ^hImm.	0.750**	1.255***	1.748***	2.202***	2.442***
	(0.294)	(0.290)	(0.327)	(0.405)	(0.663)
KP F-stat	151.6	147.3	154.6	155.3	56.8
Panel B: Excluding top-10 municipalities by 2014 population					
Δ^hImm.	0.845***	1.336***	1.789***	2.278***	2.498***
	(0.276)	(0.294)	(0.340)	(0.425)	(0.715)
KP F-stat	148.8	142.9	149.7	151.0	57.8
Panel C: Excluding top-20 municipalities by 2014 population					
Δ^hImm.	0.694**	1.240***	1.769***	2.315***	2.768***
	(0.282)	(0.297)	(0.344)	(0.433)	(0.675)
KP F-stat	137.0	132.0	141.4	143.8	52.2
Panel D: Unweighted regressions					
Δ^hImm.	0.889***	1.525***	1.989***	2.662***	3.347***
	(0.286)	(0.305)	(0.319)	(0.374)	(0.808)
KP F-stat	171.6	166.8	191.0	171.5	41.4
Panel E: Fixed 2014 denominator for immigration rate					
Δ^hImm.	0.724***	1.213***	1.755***	2.089***	2.246***
	(0.253)	(0.263)	(0.311)	(0.391)	(0.790)
KP F-stat	211.6	206.4	203.9	170.0	75.8

Δ^hImm. instrumented with the corresponding h-period shift-share predictor. Panel E fixes the population denominator at its 2014 value so that changes in the immigration rate reflect exclusively changes in the immigrant stock. All specifications include province FE, province trends (except long diff.), demand and supply controls, the exogenous spatial pull, and are weighted by 2014 population (except Panel D). SE clustered at municipal level in (1)–(4), province level in (5). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Heterogeneity by Urban Degree: Full Estimates

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Δ^h log Avg. Price	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	*2014–2025* (5)
Δ^hImm. (Urban)	0.547*	0.887***	1.263***	1.270***	0.993
	(0.290)	(0.280)	(0.329)	(0.414)	(0.714)
Δ^hImm.$\times$ Non-Urban	0.200	0.563**	0.970***	1.674***	3.087***
	(0.281)	(0.261)	(0.275)	(0.348)	(0.533)
KP F-stat	131.2	116.1	112.1	94.1	17.7
SW-F: Δ^hImm.	218.1	200.4	184.2	148.3	75.4
SW-F: Interaction	338.2	296.9	252.2	205.3	23.5
N	7,689	6,990	6,291	5,592	699

All interactions instrumented with the corresponding interaction of the shift-share predictor. Full controls, province FE, province trends (except long diff.), exogenous spatial pull. Standard errors clustered at municipal level in (1)–(4), province level in (5). Weighted by 2014 population. $^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$.

Heterogeneity: Urban Hierarchy (SR)

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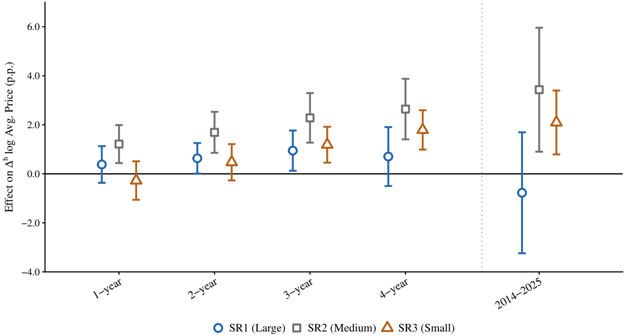
Figure: Dynamic Causal Effects of Immigration on Housing Prices by Urban Hierarchy



- Short-run response concentrated in medium-tier (SR2) municipalities; the SR1 within-group response **dissipates** beyond $h = 4$.
- Total long-run effects: SR2 $\approx$ **3.43pp**, SR3 $\approx$ **2.10pp**.

Heterogeneity by Urban Hierarchy: Full Estimates

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Δ^h log Avg. Price	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	*2014–2025* (5)
Δ^hImm. SR1 (Large)	0.383	0.635**	0.947**	0.705	−0.773
	(0.382)	(0.317)	(0.419)	(0.613)	(1.259)
Δ^hImm.$\times$ SR2 (Medium)	0.832**	1.058***	1.338***	1.937***	4.206**
	(0.342)	(0.268)	(0.328)	(0.538)	(1.775)
Δ^hImm.$\times$ SR3 (Small)	−0.656	−0.161	0.242	1.086*	2.870**
	(0.492)	(0.418)	(0.420)	(0.625)	(1.336)
KP F-stat	60.1	55.1	51.2	40.4	11.4
N	7,689	6,990	6,291	5,592	699

SR1 as omitted base category; interactions report differential effects, instrumented with the corresponding interaction of the shift-share predictor. Full controls, province FE, province trends (except long diff.), exogenous spatial pull. Standard errors clustered at municipal level in (1)–(4), province level in (5). Weighted by 2014 population. $^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$.

Native Mobility by Urban Hierarchy

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Δ^h Native Mobility	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	*2014–2025* (5)
Δ^hImm. (SR1)	0.274**	0.251**	0.206	0.081	−0.157
	(0.116)	(0.122)	(0.131)	(0.141)	(0.248)
× SR2 (Medium)	−0.139	−0.088	−0.009	0.161	0.578
	(0.092)	(0.093)	(0.098)	(0.106)	(0.389)
× SR3 (Small)	−0.115	−0.063	0.023	0.219*	0.603**
	(0.110)	(0.115)	(0.123)	(0.132)	(0.253)
KP F-stat	60.1	55.1	51.2	40.4	11.4
N	7,689	6,990	6,291	5,592	699

- Large cores (SR1) attract natives only at short horizons; in small municipalities (SR3) the differential response is positive and significant in the long run.

Mechanisms: Local Economy by Urban Degree

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	A: SS Affiliation		B: Total firms		C: Service firms		D: Income	
	Urban	Non-urb.	Urban	Non-urb.	Urban	Non-urb.	Urban	Non-urb.
1 year	0.318	-0.626^{***}	0.215^{*}	0.462^{***}	-0.044	0.521^{***}	0.260^{**}	0.608^{***}
	(0.192)	**(0.170)**	**(0.124)**	**(0.096)**	**(0.171)**	**(0.130)**	**(0.117)**	**(0.145)**
2 years	0.316	-0.471^{***}	0.367^{***}	0.419^{***}	0.049	0.480^{***}	0.989^{***}	0.824^{***}
	(0.200)	**(0.178)**	**(0.107)**	**(0.100)**	**(0.155)**	**(0.134)**	**(0.231)**	**(0.278)**
3 years	0.299	-0.273	0.488^{***}	0.467^{***}	0.097	0.551^{***}		
	(0.225)	**(0.204)**	**(0.115)**	**(0.118)**	**(0.166)**	**(0.150)**		
4 years	0.261	-0.114	0.651^{***}	0.511^{***}	0.158	0.630^{***}		
	(0.280)	**(0.266)**	**(0.134)**	**(0.144)**	**(0.188)**	**(0.180)**		
2014–2025	0.456	0.289	0.892^{***}	0.940^{***}	0.027	1.177^{***}	1.198^{***}	0.348
	(0.499)	**(0.645)**	**(0.200)**	**(0.356)**	**(0.251)**	**(0.244)**	**(0.447)**	**(0.495)**

Supply-side results (Table C.2, Panels F–I): within non-urban municipalities, **available urban land turns significantly negative in the long run** (-1.858^{**}) while the **hospitality-share response is positive and significant** across horizons (e.g., 2.973^{**} at $h=1$); residential shares remain unchanged in both groups — suggesting building activity is directed toward tourist rather than residential uses.

Within-group 2SLS effects (Table C.2 of the paper). Weighted by 2014 population. Clustered SE at municipal level (overlapping), province level (long diff.). Income long diff. covers 2020–2023. $^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$.

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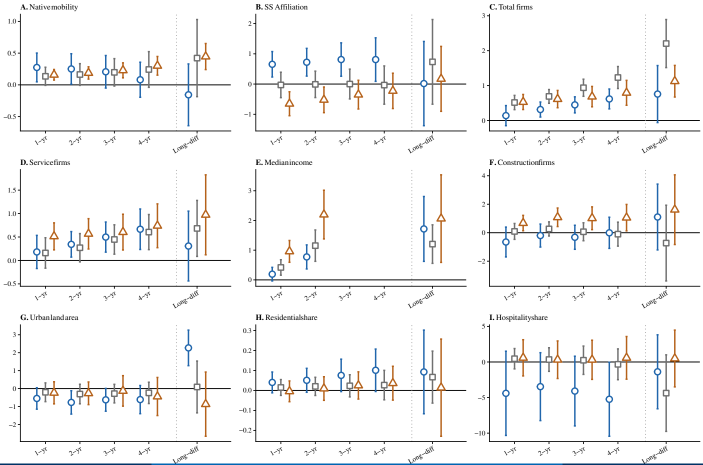
Figure: The Effect of Immigration on Other Outcomes by Urban Hierarchy



Mechanisms: Local Economy by Urban Hierarchy

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	A: SS Affiliation			B: Total firms			C: Service firms			D: Income		
	SR1	×SR2	×SR3	SR1	×SR2	×SR3	SR1	×SR2	×SR3	SR1	×SR2	×SR3
1 year	0.654***	−0.683***	−1.305***	0.140	0.376***	0.390***	0.181	−0.022	0.333*	−1.052***	0.945***	0.153
2 years	0.722***	−0.731***	−1.243***	0.314***	0.376***	0.300**	0.343**	−0.072	0.225	−0.773***	0.912***	0.189
3 years	0.811***	−0.810***	−1.161***	0.448***	0.490***	0.236	0.498***	−0.051	0.106	—	—	—
4 years	0.810**	−0.843***	−1.035***	0.618***	0.614***	0.176	0.666***	−0.061	0.074	—	—	—
2014–2025	0.017	0.718	0.155	0.758*	1.447***	0.368	0.307	0.376	0.666	1.109**	0.381	0.821

2SLS with SR1 as base category; interactions report differential effects (Table C.5 of the paper; standard errors omitted for space — see paper). Income long diff. covers 2020–2023. Weighted by 2014 population. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.